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Dynamic comb for conveyance systems for moving people or objects

Abstract

A dynamic comb for use with a conveyance system to transition a passenger or object from a nonmoving element to a moving element and vice versa. The dynamic comb can include a upper support element or a lower support element. The dynamic comb can also include a roller element disposed at least partially between and below portions of the upper support element. A method of installing the dynamic comb for an existing conveyance system or as part of a new conveyance system to transition a passenger or object from a nonmoving element to a nonmoving element and vice versa.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a conversion of U.S. Provisional application having U.S. Ser. No. 63/561,835, filed Mar. 6, 2024, which claims the benefit under 35 U.S.C. 119(e). The disclosure of which is hereby expressly incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

1. Field of the Invention

(1) The present disclosure relates to a dynamic comb for use with a conveyance system to prevent the entrapment of objects between moving and nonmoving elements of people or objects.

2. Description of the Related Art

(2) Typical conveyance systems, such as moving walkways and escalators, include a static comb positioned between nonmoving elements and moving elements. The static combs create pinch points where objects can be caught. When objects are caught at these pinch points there can be

damage to the conveyance system and/or injuries to people or damage to objects. Either one can cause the conveyance system to be shut down and not operational. There is also a height change between the moving elements of the conveyance system and the nonmoving elements of the conveyance system. This height change, combined with the static nature of the static comb, can also cause other problems for people or objects trying to use the conveyance system.

(3) Accordingly, there is a need for a dynamic comb that can prevent entrapment of objects between moving and nonmoving elements of a conveyance system and allow for easier transition between the moving and nonmoving elements of the conveyance system.

SUMMARY OF THE DISCLOSURE

(4) The present disclosure is directed to a dynamic comb for use with a conveyance system to transition a passenger or object from a nonmoving element to a moving element and vice versa. The dynamic comb can include an upper support element or a lower support element. The dynamic comb can also include a roller element disposed at least partially between and below portions of the upper support element.

(5) The present disclosure is directed to a method of installing the dynamic comb for an existing conveyance system or as part of a new conveyance system to transition a passenger or object from a nonmoving element to a nonmoving element and vice versa.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of one type of conveyance system using a dynamic comb constructed in accordance with the present disclosure.

(2) FIG. 2 is a perspective views of another type of conveyance system implementing a dynamic comb constructed in accordance with the present disclosure.

(3) FIG. 3 is a close-up, perspective view of the dynamic comb in use with a conveyance system constructed in accordance with the present disclosure.

(4) FIG. 4A is a perspective view of the dynamic comb constructed in accordance with the present disclosure.

(5) FIG. 4B is a close-up, perspective view of the dynamic comb shown in FIG. 4A constructed in accordance with the present disclosure.

(6) FIGS. 5A-5C provide multiple views of parts of the dynamic comb constructed in accordance with the present disclosure.

(7) FIGS. 6A-6C are various views of certain aspects of the dynamic comb constructed in accordance with the present disclosure.

(8) FIG. 7 is a partial side elevation view and partial cross-sectional view of a portion of the dynamic comb constructed in accordance with the present disclosure.

(9) FIG. 8 is a perspective view of another part of the dynamic comb constructed in accordance with the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

(10) Referring now to FIGS. 1-3, the present disclosure relates to a dynamic comb **10** for improving the transition between moving elements **12** and nonmoving elements **14** of a conveyance system **16**, such as a moving walkway **16a**, an escalator or a moving hand support system **16b** used with a moving walkway or escalator. The conveyance system **16** could also be an apparatus used for moving objects. The moving elements **12** may be the belt of a moving walkway **16a** or a moving hand support system **16b**, or the steps of an escalator. The nonmoving elements **14** are the areas immediately adjacent to the dynamic combs **10** on the opposite side of the moving elements **12**. The dynamic comb **10** can be placed at the beginning or end of a conveyance system **16**. The improvements in the transition can include the prevention of objects, limbs of humans or pets,

and/or other materials being entrapped between the moving and nonmoving elements **12** and **14** of the conveyance system **16** and easier transition of things, such as wheelchairs, between the moving and nonmoving elements **12** and **14** due to any height difference therebetween. The easier transition is accomplished by the dynamic comb **10** being more flushly mounted with the moving elements **12** of the conveyance system **16**. In other words, the parts of the dynamic comb **10** can be lower (or not higher) than the moving elements **12** of the conveyance system **16**, or the dynamic comb **10** is only slightly higher than the moving elements **12** of the conveyance system **16**. The present disclosure is also directed to a method of operating or installing a conveyance system **16** that includes a dynamic comb **10** as described herein, or installing a dynamic comb **10** into a conveyance system **16**.

(11) Referring now to FIGS. 4A-6C, the dynamic comb **10** can include a roller element **18**, a lower support element **20** and an upper support element **22**. The dynamic comb **10** can be designed to work with the roller element **18** and only the lower support element **20**, or it could be designed to work with the roller element **18** and only the upper support element **22**. In one embodiment, the dynamic comb **10** can be made up of multiple comb sections **24** wherein each comb section **24** includes a roller element **18**, a lower support element **20** and an upper support element **22**. The use of multiple comb sections **24** allows for easy replacement of one of the sections **24** of the dynamic comb **10** if a particular comb section **24** gets damaged and the multiple sections can also prevent excessive deflection when the dynamic comb **10** is implemented for wider systems. The dynamic comb **10** can also include a base structure **26** that supports the lower support element **20**, the upper support element **22** and the roller element **18**. The lower support element **20** and the upper support element **22** are static structural components that support and protect the roller element **18** that rotates between the upper and lower support elements **20** and **22** and parts thereof. The roller elements **18** are set up to rotate in the same direction as the direction the moving elements **12** of the conveyance system **16** moves and at roughly the same speed as the moving elements **12** of the conveyance system **16**. In certain scenarios, the roller elements **18** can be set up to rotate at a speed slower or faster than the speed of the adjacent moving element **12**.

(12) Each roller element **18**, as shown in FIG. 5A, can include a shaft member **28** with a plurality of disc members **30** extending therefrom along the length of the shaft **28**. The shaft **28** of the roller element **18** can be rotated, which rotates the disc elements **30** attached thereto. The space between each disc member **30** can be sized to accept various parts of the upper and lower support elements **20** and **22**.

(13) The lower support element **20**, as shown in FIG. 5B, can have a base portion **32** that is supported by the base structure **26** on a bottom side **34** and a plurality of fin elements **36** extending from the base portion **32**. The spaces between the fin elements **36** are sized such that the disc members **30** of the roller element **18** of the dynamic comb **10** can freely rotate therebetween, but not so big that there is space created where objects could become lodged in the spaces. Each fin element **36** of the lower support element **20** can include an arc-shaped cutout **38** to permit at least part of the shaft **28** of the roller element **18** to extend through the dynamic comb **10**. The ends **40** of the lower support elements **20** can include a flange member **42** for supporting a bearing device **44** that engages with the shaft **28** of the roller element **18**. The lower support **20** can also include a means for being secured to the upper support element **22**. The two parts can be secured to one another via any manner known to one of ordinary skill in the art. In one exemplary embodiment, the lower support element **20** could have threaded openings **46** therein for accepting screws that extend through the upper support element **22** to secure the two parts together. The lower support element **20** can be secured to the base structure **26** of the dynamic comb **10** via any manner known in the art.

(14) The upper support element **22**, as shown in FIG. 5C, can have a body portion **48** having a plurality of fin elements **50** extending therefrom. Similar to the lower support element **20**, the spaces between the fin elements **50** are sized such that the disc members **30** of the roller element **18**

of the dynamic comb **10** can freely rotate therebetween, but not be so big that objects can become lodged in the spaces. Each fin element **50** of the upper support element **22** can include an arc-shaped cutout **52** to permit at least part of the shaft **28** of the roller element **18** to extend through the dynamic comb **10**. The ends **54** of the upper support elements **22** can include a flange member **56** for working with the flange **42** of the lower support member **20** to support the bearing device **44** that engages with the shaft **28** of the roller element **18**. The body portion **48** of the upper support element **22** can be positioned on an upper side **58** of the base portion **32** of the lower support element **20**. The top sides **60** of the fin elements **50** of the upper support element **22** are curved to facilitate the transition from between the moving and nonmoving elements **12** and **14** of the conveyance system **16**.

(15) Referring now to FIG. 7, the dynamic comb **10** can also include a driving apparatus **62** for providing the rotation to the shaft **28** of the roller element **18** to rotate the shaft **28** and the disc members **30**. The driving apparatus **62** can include a motor **64** that is dedicated to the dynamic comb **10**, or it could be a motor that is used to drive other parts of the conveyance system **16**. The shaft **28** of the roller element **18** could extend and engage directly with the motor **64**. In another embodiment, the shaft **28** could include a gear **66** on one end that engages with a motor gear **68** operationally supported by the motor **64** wherein the motor **64** provides rotational movement to the motor gear **68**. Additional gears and devices can be used to facilitate proper operation of the dynamic comb **10**. In a further embodiment, the roller element **18** can be rotated by a pulley and belt-driven mechanism. The motor can have a gearbox or similar drive connection for the purposes of optimizing speed, torque, power efficiency, dimensional packaging, or another factor. The motor can be intelligently controlled by a computing device so that the speed of the roller element is very close to, or the same as, the speed of the moving elements **12** of the conveyance system **16**.

(16) When the dynamic comb **10** incorporates multiple sections **24**, the rotation of a primary roller element **18a** has to be transferred to the other roller elements **18** of the dynamic comb **10**. To transfer the rotation of one roller element **18** to another, the ends **70** of the roller elements **18** can include a power-transferring mating configuration, such as a spline or half circle. A half circle **72** on the end **70** of a roller **18** is shown in FIG. 8. The half circle **72** of one roller element **18** could be engaged with another half circle **72** on the end **70** of an adjacent roller element **18** to transfer rotation. In another embodiment, one end of the roller element could include a protrusion and the end of the adjacent roller element could have a recessed part wherein the protrusion has a given cross-sectional shape that mates with a corresponding shape of the recessed part such that movement of one roller element forces movement of the adjacent roller element. For example, the cross-sectional shape of the protrusion can be star shaped, or the protrusion could have any polygonal shape. The bearing devices **44** can be oriented such that the ends **70** of two shafts **28** of adjacent roller elements **18** can each extend at least partially into and be supported by the bearing device **44**.

(17) The disc members **30** of the roller element **18** can extend slightly beyond the fin elements **50** (more specifically the curved outer sides **60** of the fin elements **50**) of the upper support element **22** so that the outer edge **74** of the disc members **30** is the closest part of the dynamic comb **10** to the moving elements **12** of the conveyance system **16**. The distance between the disc members **30** and the moving elements **12** of the conveyance system **16** are minimized to help prevent objects from being able to get in between the dynamic comb **10** and the moving elements **12** of the conveyance system **16**. The dynamic comb **10** is designed such that the width of the disc members **30** is greater than the width of the spaces between the disc members **30**, and thus, the total width of the combined disc members' width is greater than the total width of the of the combined width of the spaces between the disc members **30**. The greater width of the disc members **30** and the rotation of the disc members **30** contributes to the ejection of objects that might find their way in the area where the dynamic comb **10** is closest to the moving elements **12** of the conveyance system **16**.

(18) From the above description, it is clear that the present disclosure is well-adapted to carry out

the objectives and to attain the advantages mentioned herein as well as those inherent in the disclosure. While presently preferred embodiments have been described herein, it will be understood that numerous changes may be made which will readily suggest themselves to those skilled in the art and which are accomplished within the spirit of the disclosure and claims.

Claims

1. A dynamic comb for use with a conveyance system to transition a passenger or object from a nonmoving element to a moving element and vice versa, the dynamic comb comprising: an upper support element; a roller element disposed at least partially between and below portions of the upper support element; wherein the dynamic comb includes multiple comb sections that cooperate to make up the dynamic comb, each comb section includes a separate roller element and upper support element, the movement of each roller element of each comb section is transferred to adjacent roller elements.
2. The dynamic comb of claim 1 wherein the roller element includes disc members extending from a shaft.
3. The dynamic comb of claim 2 wherein the upper support element includes first fin elements the disc members extend and rotate between.
4. The dynamic comb of claim 3 wherein the first fin elements include arc-shaped cutouts that permit the shaft of the roller element to extend through a portion of the upper support element.
5. The dynamic comb of claim 3 wherein outer edges of the disc members are about the same height as the height of top sides of the first fin elements.
6. The dynamic comb of claim 3 wherein outer edges of the disc members are disposed above the height of top sides of the first fin elements.
7. The dynamic comb of claim 1 further comprising a lower support element that cooperates with the upper support element to support the roller element.
8. The dynamic comb of claim 1 further comprising a driving apparatus to provide rotational movement to the roller element in a direction that matches the direction the moving element of the conveyance system travels.
9. A method, the method comprising: installing a dynamic comb for an existing conveyance system or as part of a new conveyance system to transition a passenger or object from a nonmoving element to a nonmoving element and vice versa, the dynamic comb comprising: an upper support element; a roller element disposed at least partially between and below portions of the upper support element; wherein the dynamic comb includes multiple comb sections that cooperate to make up the dynamic comb, each comb section includes a separate roller element and upper support element, the movement of each roller element of each comb section is transferred to adjacent roller elements.
10. The method of claim 9 wherein the roller element includes disc members extending from a shaft.
11. The method of claim 10 wherein the upper support element includes first fin elements the disc members extend and rotate between.
12. The method of claim 11 wherein the first fin elements include arc-shaped cutouts that permit the shaft of the roller element to extend through a portion of the upper support element.
13. The method of claim 11 wherein outer edges of the disc members are about the same height as the height of top sides of the first fin elements.
14. The method of claim 11 wherein outer edges of the disc members are disposed above the height of top sides of the first fin elements.
15. The method of claim 9 further comprising a driving apparatus to provide rotational movement to the roller element in a direction that matches the direction the moving element of the conveyance system travels.

16. A dynamic comb for use with a conveyance system to transition a passenger or object from a nonmoving element to a moving element and vice versa, the dynamic comb comprising: a lower support element; a roller element disposed at least partially between and above portions of the lower support element; wherein the dynamic comb includes multiple comb sections that cooperate to make up the dynamic comb, each comb section includes a separate roller element and lower support element, the movement of each roller element of each comb section is transferred to adjacent roller elements.

17. The dynamic comb of claim 16 wherein the roller element includes disc members extending from a shaft.

18. The dynamic comb of claim 16 further comprising an upper support element that cooperates with the lower support element to support the roller element.

19. The dynamic comb of claim 18 wherein the upper support element includes first fin elements the disc members extend and rotate between.

20. The dynamic comb of claim 19 wherein outer edges of the disc members are about the same height as the height of top sides of the first fin elements or disposed above the height of top sides of the first fin elements.

21. The dynamic comb of claim 19 wherein the first fin elements include arc-shaped cutouts that permit the shaft of the roller element to extend through a portion of the upper support element.
